

Signature Project Summary

2026 Internship Program

ClaimLearn: Probabilistic Modeling for Claims Audit Optimization

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The ClaimLearn project is a machine learning initiative to improve the claims audit prioritization process, enabling a statistical model to identify claims most likely to contain payment errors and report them for internal audit. Currently this is done via random sampling at a ~2-3% error rate with any improvement carrying significant financial implications, both in optimizing audit workload and in raising the catch rate for erroneous claims.

PROJECT DESCRIPTION

Using ~92,000 AMISYS claim instances audited across 10 quarters (Q1 2024 – Q2 2026), a pipeline joins claim-header records to line-level AMISYS transactions under a strict temporal cutoff to prevent post-audit information leakage, then produces a claim-level feature matrix and ranks by predicted error probability.

Feature engineering yielded a 175-feature representation, including standardized dollar amounts, binary indicators, interactions, group-aware residual stacking, and risk encodings of high-cardinality variables via empirical-Bayes target encoding, together with an unsupervised anomaly score via Isolation Forest. All features were algorithmically screened for low variance and multicollinearity, with feature-family ablation and a VIF-to-SHAP filter confirming their contribution to the top-K precision.

Modeling spans unsupervised, supervised, weakly-supervised deep learning, and Bayesian probabilistic models with performance measured by precision, or the rate at which high-risk instances are true errors. The highest performing are LightGBM and XGBoost, achieving a >10× lift over the random-sampling baseline on the top 100 highest-risk flagged claims. Developed in parallel, the BART (Bayesian Additive Regression Trees) model matches performance while additionally producing calibrated posterior uncertainty estimates, enabling audit selection to be framed as constrained resource allocation under uncertainty rather than fixed-threshold classification, a direction for future work.

ClaimLearn: Uncertainty-Aware Audit Prioritization
BART on held-out test set — 18,434 claims, 449 confirmed errors (2.44% base rate)

